6A INTRODUCING ENTHALPY AND ENTHALPY CHANGE

SPECIFICATION REFERENCE
6.1 | 6.4 PART

LEARNING OBJECTIVES

- Know that the enthalpy change, ΔH, is the heat energy change measured at constant pressure and that standard conditions are 100 kPa and a specified temperature, usually 298 K.
- Be able to define what we mean by standard enthalpy of reaction.

CHEMICAL AND HEAT ENERGY

The first law of thermodynamics states that, during a chemical reaction, energy cannot be created or destroyed. However, one form of energy can be transferred into another form.

Various forms of energy are interesting to a chemist. Two of the most important ones are:

- chemical energy
- heat energy.

CHEMICAL ENERGY

Chemical energy is made up of two components:

- Kinetic energy, which is a measure of the motion of the particles (atoms, molecules or ions) in a substance.
- Potential energy, which is a measure of how strongly these particles interact with one another (i.e. both attract and repel one another).

HEAT ENERGY

Heat energy is the portion of the potential energy and the kinetic energy of a substance that is responsible for the temperature of the substance.

The heat energy of a substance is directly proportional to its absolute temperature (i.e. the temperature measured in Kelvin).

ENTHALPY AND ENTHALPY CHANGES

Enthalpy is a measure of the total energy of a system. When considering a chemical reaction, the 'system' refers to the reaction mixture. Everything outside of the system is called the 'surroundings', which in practice is the air in the room in which the reaction is taking place.

You cannot directly determine the enthalpy of a system, but you can measure the enthalpy change (ΔH) that takes place during a physical or a chemical change.

The enthalpy change of a process is the heat energy that is transferred between the system and the surroundings at *constant pressure*.

ADDITIONAL READING

Enthalpy is the sum of the internal energy of a system (the energy required to create the system) and the amount of energy required to make room for the system by displacing its environment and establishing its volume and pressure.

Enthalpy has the symbol H and internal energy has the symbol U. This equation shows their relationship:

$$H = U + pV$$

where p is the pressure of the system and V is its volume.

LEARNING TIP

Do not confuse heat energy with heating. Heating is the result of a transfer of heat energy from one system to another, which in turn produces a change of temperature.

For example, when water at 60 °C is put in contact with air at 20 °C, heat energy will be transferred from the water to the air. This will result in an increase in temperature of the air, with a subsequent decrease in temperature of the water. This is why a cup of hot coffee gets cold when you forget to drink it.

EXOTHERMIC AND ENDOTHERMIC PROCESSES AND REACTIONS

Two types of process can take place. These are:

- **exothermic** – where heat energy is transferred from the system to the surroundings
- **endothermic** – where heat energy is transferred from the surroundings to the system.

Examples of exothermic and endothermic processes are given in **table A**.

EXOTHERMIC	ENDOTHERMIC
Freezing water	Melting ice
Condensing water vapour	Evaporating water
Dissolving sodium hydroxide in water	Dissolving ammonium nitrate in water
Reaction between dilute hydrochloric acid and aqueous sodium hydroxide	Reaction between dilute ethanoic acid and solid sodium hydrogencarbonate
Combustion of petrol	Photosynthesis

table A Examples of exothermic and endothermic processes.

EXAM HINT

When water condenses, hydrogen bonds are formed between water molecules. Bond formation is an exothermic process. See **Topic 7** for an explanation of hydrogen bonds.

An example of an exothermic reaction is:

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$
$$\Delta H = -57.1\,kJ\,mol^{-1}$$

An example of an endothermic reaction is:

$$C_6H_8O_7(aq) + 3NaHCO_3(s) \rightarrow C_6H_5O_7^{3-}(aq) + 3Na^+(aq) + 3CO_2(g) + 3H_2O(l)$$
(citric acid)

$$\Delta H = +70\,kJ\,mol^{-1}$$

LEARNING TIP

The negative (–) sign indicates that heat energy is transferred from the system to the surroundings.

The positive (+) sign indicates that heat energy is transferred from the surroundings to the system.

Exothermic reactions can usually be recognised because they result in an immediate increase in temperature. For example, when hydrochloric acid is added to aqueous sodium hydroxide, the temperature of the reaction mixture increases. Similarly, when natural gas burns in oxygen, the flame produced is hot.

Conversely, endothermic reactions often produce a decrease in temperature of the reaction mixture, for example when solid sodium hydrogencarbonate is added to aqueous citric acid.

Any reaction that has to be continually heated in order for it to take place is endothermic. For instance, the thermal decomposition of calcium carbonate into calcium oxide and carbon dioxide is an endothermic reaction:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \qquad \Delta H = +178\,kJ\,mol^{-1}$$

STANDARD CONDITIONS

In 1982, the International Union of Pure and Applied Chemistry (IUPAC) recommended that all enthalpy changes should be quoted using standard conditions of 100 kPa pressure and a stated temperature. The temperature most commonly used is 298 K.

Under these conditions, the enthalpy change measured is called the 'standard enthalpy change', and is given the symbol $\Delta H^{\ominus}_{298K}$ or simply $\Delta H^{\ominus}$.

LEARNING TIP

Some data books and textbooks quote standard enthalpy changes at a standard pressure of one atmosphere, i.e. 1 atm (1 atm = 101.325 kPa). This is not in agreement with IUPAC recommendations. For this reason, we will use 100 kPa as the standard pressure for thermochemical measurements.

STANDARD ENTHALPY CHANGE OF REACTION, $\Delta_r H^{\ominus}$

When looking at **standard enthalpy change of reaction** ($\Delta_r H^{\ominus}$), it is important to recognise that the enthalpy change is for the reaction *as written*.

For the reaction

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g) \qquad \Delta_r H^{\ominus} = -92\,kJ\,mol^{-1}$$

But for the reaction when written as:

$$\tfrac{1}{2}N_2(g) + 1\tfrac{1}{2}H_2(g) \rightarrow NH_3(g) \qquad \Delta_r H^{\ominus} = -46\,kJ\,mol^{-1}$$

In each case the 'per mole' refers to *one mole of equation*, and not to one mole of any reactant or product.

EXAM HINT

Be sure to include the correct state symbols when writing an equation for standard enthalpy change.

CHECKPOINT

1. Classify each of the following as exothermic or endothermic processes:
   (a) cooking an egg
   (b) formation of snow in clouds
   (c) burning candle wax
   (d) forming a cation from an atom in the gas phase
   (e) baking bread.

2. Sherbet is a solid mixture of sodium hydrogencarbonate and citric acid. If you put sherbet in your mouth and mix it with saliva, your mouth will feel cold. Explain why.

SUBJECT VOCABULARY

exothermic a reaction where heat energy is transferred from the system to the surroundings

endothermic a reaction where heat energy is transferred from the surroundings to the system

standard enthalpy change of reaction the enthalpy change which occurs when equation quantities of materials react under standard conditions